De La Salle University - Dasmariñas
COLLEGE OF INFORMATION AND COMPUTER STUDIES
INFORMATION TECHNOLOGY DEPARTMENT

COURSE SYLLABUS

First Semester, SY 2024-2025

COURSE CODE	**:**	**S-ITNT415Lc**
COURSE TITLE	**:**	**S-ITNT415Lc – Cisco Networking Semester 4 (DevNet)**
COURSE TYPE	**:**	**Lecture**
COURSE CREDIT	**:**	**Two (2) units**
LEARNING MODALITY	**:**	**Flexible Blended Learning ModelTNT344L**
PRE-REQUISITES	**:**	
CO-REQUISITES	**:**	**S-ITNT415Lc**
PROFESSOR	**:**	**Maria Gloria Del Rosario**
CONSULTATION HOURS	**:**	**Thursday 10:00-12:00, 1:00-3:00 COS100C**
CONTACT INFORMATON:		**madelrosario@dlsud.edu.ph (046) 4811900 local 3134**

VISION

A leading Catholic university that inspires excellence and drives innovation toward a just, peaceful, and sustainable society.

MISSION

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension programs.

COURSE DESCRIPTION:

Discusses the WAN technologies and network services required by converged applications in a complex network. The course enables students to understand the selection criteria of network devices and WAN technologies to meet network requirements. Students learn how to configure and troubleshoot network devices and resolve common issues with data link protocols. Students also develop the knowledge and skills needed to implement virtual private network (VPN) operations in a complex network. The course gears towards DevNet Associate certification exam. In addition to this, DevNet Associate course teaches students the best practices of DevOps,

COURSE LEARNING OUTCOME

By the end of this course, students can:

CLO1. develop basic Python programming and Linux skills.
CLO2. investigate development environment using DevNet resources.
CLO3. use software development and design best practices.
CLO4. create a secure REST API.
CLO5. explain the processes and devices that support network connectivity.
CLO6. use current technologies to deploy and secure applications and data in a cloud environment.
CLO7. analyze software testing and deployment methods in automation and simulation environments.
CLO8. analyze Cisco platforms used for collaboration, infrastructure management, and automation.

LEARNING PLAN:

Learning Outcomes for Midterm Period

Course Learning Outcomes	**Topic Learning Outcomes**
Midterm Period	
CLO1. Develop basic Python programming and Linux skills.	TLO1. Manage the Linux file systems and permissions TLO2. Use basic programming
CLO2. Investigate a development environment using DevNet resources.	TLO3. Investigate DevNet online resources
CLO3. Use software development and design best practices.	TLO4. Describe the benefits of various software design patterns
CLO4. Create a secure REST API	TLO5. Compare synchronous and asynchronous API design styles.

Module Number and Learning Outcome(s)	Week	Teaching-Learning Activities / Assessment Strategy	Technology Enabler	A nchronou	Synchronous T	e Alloted
Module 1 CLO 1 TLO 1	1 Aug 19-24	Gospel Reading, Reflection and Prayer Presentation on the Fully online learning model LMS: 1. Schoolbook enlistment 2. MS Teams channel for every synch meeting 3. Netacad Course Experience – Enrollment, Interface, Submission, Getting Used to the Cisco Networking Academy LMS Module 1: Setup Your Lab Environment Objective: Prepare a Computer for Virtualization Present the course overview Define the new key terms in this module Class Participation 1: Handwritten Activity on Module 1 Key Terms and Concepts	MS Teams Schoolbook NetAcad		/	2
Module 1 CLO 1 TLO 2	2 Aug 26 - 31	Gospel Reading, Reflection and Prayer Review of Linux Commands Module 1: Setup Your Lab Environment Objective: Use basic Python programming Formative Activity: Watch Video on The Power of Code, Practice Python Programming Enabling Activity 1: Module 1 Exam via netacad	MS Teams NetAcad		/	2
Module 2 CLO 2 TLO 3	3 Sep 2-7	Gospel Reading, Reflection and Prayer Module 2: The DevNet Developer Environment Objective: Explore DevNet Online Resources Formative Activity: Visit the DevNet Home page at developer.cisco.com and showcase the DevNet online resources to the class. Class Participation 2: Handwritten Activity on Module 2 Key Terms and Concepts	MS Teams NetAcad	/		2

Module 2 CLO 2 TLO 3	4 Sep 9-14	Gospel Reading, Reflection and Prayer Module 2: The DevNet Developer Environment Objective: Explain how DevNet encourages communities of network programmers. Explore: Getting started with DevNet Resources Enabling Activity 2: Module 2 Exam in netacad	MS Teams NetAcad		/	2
Module 3 CLO 3 TLO 4	5 Sep 16-21	Gospel Reading, Reflection and Prayer Module 3: Software Development and Design Objective: Use software development and design best practices Discussion on Git and GitHub Class Participation 3: Handwritten Activity on Module 2 Key Terms and Concepts	MS Teams NetAcad	/		
Module 3 CLO 3 TLO 4	6 Sep 23-28	Gospel Reading, Reflection and Prayer Module 3: Software Development and Design Objective: Use software development and design best practices Compare software development methodologies Describe the benefits of various software design Compare Waterfall, Agile, and Lean Code review Enabling Activity 3: Module 3 Exam in netacad	MS Teams NetAcad		/	
Module 3 CLO 3 TLO 4	7 Sep 30 – Oct 5	Gospel Reading, Reflection and Prayer Module 3: Software Development and Design Objective: Use software development and design best practices Review Git and GitHub Enabling Activity 4: Software Version Control with Git	MS Teams NetAcad	/		2
Module 3 CLO 3 TLO 4	8 Oct 7-12	Gospel Reading, Reflection and Prayer Module 3: Software Development and Design Objective: Use software development and design best practices Review notes on Git and GitHub Formative Activity: Software Development Scenarios Enabling Activity 5: Explore Python Classes	MS Teams NetAcad	/		2
Modules 1,2,3	9 Oct 14-19	Summative Assessment baed on Module 1-3	MS Team Schoolbook NetAcad		/	
			TOTAL	8	10	18

LEARNING PLAN FOR FINALS:

Course Learning Outcomes	Topic Learning Outcomes
Final Period	
CLO5. Explain the processes and devices that support network connectivity	TLO6. Apply the processes and devices that support network connectivity
CLO6. Use current technologies to deploy and secure applications and data in a cloud environment.	TLO7. Explain the use of Continuous Integration/Continuous Deployment (CI/CD) in application deployment
CLO7. Analyze software testing and deployment methods in automation and simulation environments.	TLO8. Describe deployment environments that benefit from automation
CLO8. Analyze Cisco platforms used for collaboration, infrastructure management, and automation.	TLO9. Compare network programmability models

Learning Outcomes for Final Period

Module Number and Learning Outcome(s)	Week	Teaching-Learning Activities / Assessment Strategy	Technology Enabler	Asynchronous	Onsite/Face to Face	Allotted Hours
Module 4 CLO 4 TLO 2	10 Oct 21-26	Gospel Reading, Reflection and Prayer Module 4: Understanding and Using APIs Objective: Create REST API requests over HTTPS to securely integrate services. Class Participation 1: Handwritten Activity on Module 4 and 5 Key Terms and Concepts	MS Teams NetAcad		/	2
Module 4 CLO 4 TLO 2	11 Oct 28-Nov2	Independent Learning Week				
Module 4 CLO 4 TLO 2	12 Nov 4-9	Gospel Reading, Reflection and Prayer Module 5: Network Fundamentals Objective: Apply the processes and devices that support network connectivity Enabling Activity 1: Module 4 & 5 exam in netacad	MS Teams NetAcad			

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Module 5 CLO 4 TLO 2	13 Nov 11-16	Gospel Reading, Reflection and Prayer Module 5: Network Fundamentals Objective: Apply the processes and devices that support network connectivity Enabling Activity 2: Module 6 exam in netacad	MS Teams NetAcad	/		2
Module 5 CLO 2 TLO 6	14 Nov 18-23	Gospel Reading, Reflection and Prayer Module 5: Network Fundamentals Objective: Apply the processes and devices that support network connectivity Discussion Explain basic network terms and processes Class Participation 2: Handwritten Activity on Module 6 Key Terms and Concepts	MS Teams NetAcad		/	2
Module 6 CLO 6 TLO 7	15 Nov 25-30	Gospel Reading, Reflection and Prayer Module 6: Application Deployment and Security Objective: Use current technologies to deploy and secure applications and data in a cloud environment Enabling Activity 3: Module 7 exam in netacad	MS Teams NetAcad	/		2
Module 7 CLO 6 TLO 7	16 Dec 2-7	Gospel Reading, Reflection and Prayer Module 7: Infrastructure and Automation Objective: Explain the principles of DevOps Class Participation 3: Handwritten Activity on Module 7-8 Key Terms and Concepts	MS Teams NetAcad		/	2
Module 8 CLO 7 TLO 8	17 Dec 9-14	Gospel Reading, Reflection and Prayer Module 8: Cisco Platforms and Development Objective: Explain how Cisco SDKs assist in the development of applications. Enabling Activity 4: Module 8 exam in netacad	MS Teams NetAcad	/		2
Modules 4,5,6,7,8	18 Dec 16-21	Summative Assessment based on Module 4-8	NetAcad/SB		/	2
			TOTAL	8	10	18

FINAL PRODUCT/PERFORMANCE:

All students taking this course must accomplish the following requirements listed below in order to gain the certificate of completion for DevNet Associate.

- Complete all online modules exams with a minimum grade of 80 for all chapter exams.

- Complete the online final exam with a minimum grade of 80.
- Accomplish the teacher-made test and course feedback online.
- Pass the Hands-on Skills Lab.
- Submit the required PT activities completed to 100%.
- Demonstrate mastery on the interactive activity on key terms and concepts.

GRADING SYSTEM

Midterm

Enabling Assessments	-	50%
Class Participation	-	20%
Summative Assessments	-	30%
Total		100%

Finals

Enabling Assessments	-	50%
Class Participation	-	20%
Summative Assessments	-	30%
Total		100%

Overall Final Grade = [(Midterm grade) + (Final term grade)] / 2

COURSE POLICIES AND REQUIREMENTS:

1. This course follows the Flexible Blended Learning Mode. There will be 50% online and 50% asynchronous.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar and at the portal.

3. Communication and Feedback
    a. Online classes shall happen as indicated in the learning plan section of the syllabus. Fifty percent (50%) of the weekly classes is delivered synchronously while the remaining is asynchronous.
    b. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:
        i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
        ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through Schoolbook messaging feature.
        iii. Email – you may email your professor. (Professor's email will be given during the course orientation)
    c. No use of foul, curse, and other bad words, whether in written or verbal. No shouting. No flooding of chat and messages. No selling.

4. Attendance Policy

a. Attendance Policy for Online Classes is scheduled based on the learning plan. A maximum of 20% for class participation/recitation/attendance shall be allotted for student's grade.
b. The online meeting will only be held at the schedule indicated in the registration form.
c. Students are required to attend online classes, 40% of absence (from online classes) for the entire semester means 0.0 final grade.

5. Intellectual Property Policy
    a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
    b. Using any illegally obtained software is strictly prohibited.
    c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
    d. Giving of password to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 account other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

6. Other Policies
    a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.
    b. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.
    c. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless a special permission is sought. Cellular phones cannot also be used as calculator during examination.
    d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

REFERENCES

Accession number or eprovider	Reference Material
E-book 2655449	Singh GD. 2021. Cisco Certified CyberOps Associate 200-201 certification guide : learn blue teaming strategies and incident response techniques to mitigate cybersecurity incidents. Birmingham: Packt Publishing.
E-book 2700917	OrtegaJM. 2021. Mastering Python for Networking and Security - Second Edition Leverage the Scripts and Libraries of Python Version 3. 7 and Beyond to Overcome Networking and Security Issues. Birmingham: Packt Publishing, Limited.
e-Book QA 76.73.P98 .Or82 2019	Sarker, Dr. M. (2019). Learning Python networking : a complete guide to build and deploy strong networking capabilities using Python 3.7 and Ansible. Second edition. UK : Packt Publishing, Ltd.
E- Book 2363872	Chou E. 2020. Mastering Python networking : your one-stop solution to using Python for network automation, programmability, and DevOps. Birmingham: Packt Publishing.
E-Book 2379305	Tevault DA. 2023. Mastering Linux Security and Hardening. Packt Publishing Ltd
Website	Login | Skills for All with Cisco. 2024. Netacadcom. [accessed 2024 Aug 5]. https://lms.netacad.com/mod/lti/view.php?id=82254473.

Prepared: **INFORMATION TECHNOLOGY DEPARTMENT**
1st Semester 2024-2025

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